Data Technology Study Notes

Topic: Core Concepts in Modern Data Systems

Databases and Data Models

- **Relational databases** organize data into tables made of rows and columns. SQL is commonly used to query and modify this data.
- **Primary keys** uniquely identify rows, while **foreign keys** connect related tables.
- **NoSQL databases** are useful when data is semi-structured, rapidly changing, or does not fit a strict tabular schema.

ETL and Data Pipelines

- **ETL** stands for Extract, Transform, Load. Data is collected from sources, cleaned or reshaped, and then stored in a target system such as a data warehouse.
- A **data pipeline** automates the movement and processing of data between systems. Pipelines can run in batches or continuously in near real time.
- Data quality checks help detect missing values, duplicates, incorrect types, and inconsistent formats before data is used for analysis.

Data Warehouses and Analytics

- A **data warehouse** stores integrated historical data that is optimized for reporting and analytics rather than day-to-day transactions.
- **OLTP** systems focus on frequent inserts and updates, while **OLAP** systems focus on complex analytical queries over large amounts of data.
- Business intelligence tools such as Power BI can connect to structured data sources and turn query results into dashboards and visualizations.

Performance and Reliability

- **Indexes** can make searches and joins faster, but too many indexes can slow down inserts and updates.
- **Normalization** reduces duplicated data and improves consistency. In analytics systems, some denormalization may be used to make queries simpler and faster.
- Backups, access control, monitoring, and validation are important because useful data must also be reliable and secure.